Exercise 4 :

Model 1 :

Conditioning on y_1, we have:

$$S(\phi, \mu) = \sum_{t=2}^{n} (y_t - \mu - \phi(y_{t-1} - \mu))^2$$

to find $\hat{\mu}$ and $\hat{\phi}$, we take the partial derivatives respected to μ and ϕ, set them to 0 and solve simultaneously:

$$\frac{\partial S}{\partial \mu} = \sum_{t=2}^{n} 2[y_t - \mu - \phi(y_{t-1} - \mu)](\phi - 1) = 0$$

$$\frac{\partial S}{\partial \phi} = \sum_{t=2}^{n} 2[y_t - \mu - \phi(y_{t-1} - \mu)](\mu - y_{t-1}) = 0$$

Both equation involve $\phi \cdot \mu$ term which makes them non-linear.

Model 2 :

$$S(\phi, \theta_0) = \sum_{t=2}^{n} [y_t - \phi y_{t-1} - \theta_0]^2$$

$$\frac{\partial S}{\partial \theta_0} = -2 \sum_{t=2}^{n} [y_t - \phi y_{t-1} - \theta_0]$$

$$\frac{\partial S}{\partial \phi} = -2 \sum_{t=2}^{n} [y_t - \phi y_{t-1} - \theta_0] \cdot y_{t-1}$$

both equation are linear on θ_0 and ϕ.